

Quiz (Carolina Reaper)

- Q1.** A shipping container has a height of $2 \cdot \sqrt{3}$ meters. What is the rationalized form of its height?
- (A) $\frac{2 \cdot \sqrt{3}}{1}$
(B) $\frac{2 \cdot \sqrt{3}}{\sqrt{3}}$
(C) $2 \cdot \sqrt{3}$
(D) $\frac{2}{\sqrt{3}}$
(E) $\frac{2 \cdot \sqrt{3}}{3}$
- Q2.** A traffic flow model uses the equation $t = \frac{5}{\sqrt{x}}$ to estimate delivery times. Rationalize the denominator of this equation.
- (A) $t = \frac{5 \cdot \sqrt{x}}{x}$
(B) $t = \frac{5}{x}$
(C) $t = \frac{5 \cdot x}{\sqrt{x}}$
(D) $t = \frac{5 \cdot \sqrt{x}}{\sqrt{x}}$
(E) $t = 5 \cdot \sqrt{x}$
- Q3.** The volume of a shipping box is given by $V = \frac{1}{2} \cdot \sqrt{2} \cdot l \cdot w$. Rationalize the denominator if the length l is $4 \cdot \sqrt{3}$ meters.
- (A) $V = \frac{1}{2} \cdot \sqrt{2} \cdot 4 \cdot \sqrt{3} \cdot w$
(B) $V = \frac{1}{2} \cdot \frac{4 \cdot \sqrt{3}}{\sqrt{2}} \cdot \sqrt{2} \cdot w$
(C) $V = \frac{1}{2} \cdot 4 \cdot \sqrt{3} \cdot w$
(D) $V = \frac{1}{2} \cdot \frac{4 \cdot \sqrt{3}}{\sqrt{3}} \cdot w$
(E) $V = \frac{1}{2} \cdot 4 \cdot w$
- Q4.** The time it takes for a package to be delivered is given by $t = \frac{200}{\sqrt{x+2}}$. Rationalize the denominator.
- (A) $t = \frac{200(\sqrt{x+2})}{(\sqrt{x+2})(\sqrt{x+2})}$
(B) $t = \frac{200(\sqrt{x+2})}{(\sqrt{x-2})(\sqrt{x+2})}$
(C) $t = \frac{200}{\sqrt{x-2}}$
(D) $t = \frac{200}{\sqrt{x+2}}$
- (E) $t = 200(\sqrt{x-2})$
- Q5.** A logistics company uses the equation $d = \frac{500}{\sqrt{t}}$ to estimate the distance traveled. Rationalize the denominator.
- (A) $d = 500 \cdot \sqrt{t}$
(B) $d = \frac{500 \cdot \sqrt{t}}{t}$
(C) $d = \frac{500}{t}$
(D) $d = 500 \cdot t$
(E) $d = \frac{500 \cdot t}{\sqrt{t}}$
- Q6.** The cost of shipping a package is given by $c = \frac{10}{\sqrt{x-1}}$. Rationalize the denominator.
- (A) $c = \frac{10(\sqrt{x+1})}{(\sqrt{x-1})(\sqrt{x+1})}$
(B) $c = \frac{10(\sqrt{x-1})}{(\sqrt{x+1})(\sqrt{x-1})}$
(C) $c = \frac{10}{\sqrt{x+1}}$
(D) $c = \frac{10}{\sqrt{x-1}}$
(E) $c = 10(\sqrt{x+1})$
- Q7.** A traffic model uses the equation $v = \frac{50}{\sqrt{d}}$ to estimate the velocity of cars. Rationalize the denominator.
- (A) $v = 50 \cdot \sqrt{d}$
(B) $v = \frac{50 \cdot \sqrt{d}}{d}$
(C) $v = \frac{50}{d}$
(D) $v = 50 \cdot d$
(E) $v = \frac{50 \cdot d}{\sqrt{d}}$
- Q8.** The height of a bridge is given by $h = \frac{20}{\sqrt{x+3}}$. Rationalize the denominator.
- (A) $h = \frac{20(\sqrt{x-3})}{(\sqrt{x+3})(\sqrt{x-3})}$
(B) $h = \frac{20(\sqrt{x+3})}{(\sqrt{x-3})(\sqrt{x+3})}$
(C) $h = \frac{20}{\sqrt{x-3}}$
(D) $h = \frac{20}{\sqrt{x+3}}$
(E) $h = 20(\sqrt{x-3})$

Open Ended Questions (FRQ)

- Q9.** Rationalize the denominator of the equation $t = \frac{10}{\sqrt{2x-1}}$ and simplify.
- Q10.** The distance traveled by a car is given by $d = \frac{200}{\sqrt{t+2}}$. Rationalize the denominator and find the distance traveled when $t = 9$ seconds.

Chef's Tips

- Tip 1: Make sure to check the units of the answer.
- Tip 2: Rationalize the denominator by multiplying by the conjugate.
- Tip 3: Simplify the expression after rationalizing the denominator.